

```
In [1]: import pandas as pd

df = pd.read_csv('bridges.csv', na_values=['?', ''])
df.head()
```

```
Out[1]:
```

	Id	river	location	erected	purpose	length	lanes	clear- g	t-or-d	material
0	E1	M	3.0	1818	HIGHWAY	NaN	2.0	N	THROUGH	WOOD
1	E2	A	25.0	1819	HIGHWAY	1037.0	2.0	N	THROUGH	WOOD
2	E3	A	39.0	1829	AQUEDUCT	NaN	1.0	N	THROUGH	WOOD
3	E5	A	29.0	1837	HIGHWAY	1000.0	2.0	N	THROUGH	WOOD
4	E6	M	23.0	1838	HIGHWAY	NaN	2.0	N	THROUGH	WOOD

```
In [2]: print('missing:', df['length'].isna().sum(), 'of', len(df))
df['length'].describe()
```

missing: 27 of 108

```
Out[2]: count      81.000000
mean      1567.469136
std       747.491523
min       804.000000
25%      1000.000000
50%      1300.000000
75%      2000.000000
max      4558.000000
Name: length, dtype: float64
```

length assumption

Bridges can be very short or very long. I am going to assume that a bridge must be at least 20 ft to be a bridge, but not longer than 10000 ft to rule out anything unreasonable for the time.

```
In [3]: LOW, HIGH = 20, 10000
range_flags = df[(df['length'] < LOW) | (df['length'] > HIGH)]
range_flags[['Id', 'river', 'erected', 'purpose', 'length', 'material', 'span', 'type']]
```

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Out[3]:
```

	Id	river	erected	purpose	length	material	span	type
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Nothing got flagged.

A fixed range didn't catch anything. I'll assume a bridge length is implausible if it falls outside 1.5×IQR from the middle 50% of values (the standard statistical outlier

definition).

```
In [4]: q1, q3 = df['length'].quantile([0.25, 0.75])
iqr = q3 - q1
lo_fence, hi_fence = q1 - 1.5 * iqr, q3 + 1.5 * iqr
print(f'Q1={q1}, Q3={q3}, IQR={iqr}')
print(f'fences: [{lo_fence}, {hi_fence}]')

iqr_flags = df[(df['length'] < lo_fence) | (df['length'] > hi_fence)]
iqr_flags[['Id', 'river', 'erected', 'purpose', 'length', 'material', 'span']]
```

Q1=1000.0, Q3=2000.0, IQR=1000.0
fences: [-500.0, 3500.0]

Out[4]:

	Id	river	erected	purpose	length	material	span	type
31	E34	O	1888	RR	4558.0	STEEL	LONG	SIMPLE-T
44	E46	A	1897	RR	4000.0	STEEL	LONG	SIMPLE-T
104	E91	O	1975	HIGHWAY	3756.0	STEEL	LONG	ARCH

Anything more than $1.5 \times \text{IQR}$ below Q1 or above Q3 got flagged as a statistical outlier.

There are only 3 outliers. I will leave them because, while they are longer, they are all plausible lengths. E91 is only 256 feet from the cutoff. It seems fine. E34 and E46 are on the longer side for sure given when they were made, but seem very possible.